SUBSEA MATTRESS VERIFICATION REPORT: CR-A1_M15 (#A046)

Project: Tung-Hsiao Power Plant 2nd Stage Renewal Subsea Gas Pipeline Project
Client: Taiwan Power Company (TPC) | Main Contractor: NMDC Energy | Vessel: DP2 MMA Prestige
Crossing Array ID: CR-A1_L1 | Design Location ID: CR-A1_M15 | Mattress Tag ID: #A046
Survey Date/Time: 2026-11-04 14:34 | Deliverable Base Filename: 20261104_Mattress_CR-A1_M15_A046

1. EXECUTIVE VERDICT

Status: REVIEW / REPOSITION ADVISORY - CHECK EDGE OVERHANG & OFFSET
- Points Inside Design Footprint (6.00 m x 3.00 m): 85.6% (700 valid points; 0 outliers filtered)
- Cloud MRR Box Radial Offset: 0.265 m (dAlong: +0.050 m, dCross: -0.260 m)
- Heading Alignment Delta (dAz): +1.035 deg (Design South Brg: 195.854 deg vs. Cloud South Brg: 196.889 deg)

2. GEODETIC & DIMENSIONAL COMPARISON TABLE

Parameter	Design Baseline	Cloud MRR Box	Cloud Mean Centroid	Delta (Act - Des)
Easting E (m, TWD97)	194541.443	194541.206	194541.206	-0.237 m
Northing N (m, TWD97)	2702613.394	2702613.513	2702613.513	+0.119 m
Elev / Depth Z (TWVD2001)	Seabed + 0.30 m	-18.20	-18.20	-18.20..-18.20 m
South-Ref Bearing (deg)	195.854 deg	196.889 deg	196.889 deg	+1.035 deg
North Long-Axis Az (deg)	15.854 deg	16.889 deg	16.889 deg	+1.035 deg
Footprint Dim (L x W)	6.00 m x 3.00 m	6.00 m x 3.00 m	-	dL +0.00, dW +0.00m
Radial Center Offset (m)	0.000 m	0.265 m	0.265 m	dAl +0.05, dCr -0.26m

3. EDGE OVERHANG / CLEARANCE & NEIGHBOR PROXIMITY

- NNE Edge (+3.00 m Along-Track): Overhang +0.08 m | Neighbor: None (Array End)
- SSW Edge (-3.00 m Along-Track): Inner Clearance 0.02 m | Neighbor: None (Array End)
- ESE Edge (+1.50 m Cross-Track): Inner Clearance 0.21 m | Neighbor: CR-A1_M16 (Center Dist: 3.00 m)
- WNW Edge (-1.50 m Cross-Track): Overhang +0.32 m | Neighbor: CR-A1_M14 (Center Dist: 3.00 m)

4. OPERATIONAL CRANE NUDGE RECOMMENDATION

- Required Shift (Cloud Box Center -> Design Center): Move 0.27 m toward ESE (Azimuth 116.7 deg).
- Component Deltas: dE = +0.237 m, dN = -0.119 m.

5. SURVEYOR REMARKS & QA/QC SIGN-OFF

- Mattress #A046 at location CR-A1_M15 (CR-A1_L1) verified via 4-Corner WROV / MBES Geodetic Engine.
- All coordinates referenced to TWD97 (TM2 Zone 121, EPSG:3826) and vertical datum TWVD2001.